```
var_dump(metaphone("algorithm"));
var_dump(metaphone("algo-rhythm"));

var_dump(soundex("algorithm"));
var_dump(soundex("algo-rhythm"));

Output:
string(6) "ALKROM"
string(6) "ALKROM"
string(4) "A426"
string(4) "A426"
```

The output might look a little weird; those are the Metaphone (and Soundex) keys of the input strings. Both algorithms aim to encode (English) homophones in similar ways: the example above demonstrates that "rithm" and "rhythm" have identical pronunciation, and we can see from the output that the two strings give the same results. Soundex is the older of the two algorithms, first patented in 1918 by Robert C. Russell and Margaret King Odell; PHP's implementation follows that described in *The Art of Computer Programming, Volume 3* (Donald Knuth, 1973). Metaphone was developed from Soundex and first published in *Computer Language, Vol. 7, No. 12* (Lawrence Philips, Dec 1990). Both algorithms were created for English language strings and do not handle UTF-8 characters:

```
var_dump(soundex("L"));
var_dump(soundex("\u{141}"));
var_dump(metaphone("L"));
var_dump(metaphone("\u{141}"));

Output:
string(4) "L000"
string(4) "0000"
string(1) "L"
string(0) ""
```

A detailed description of these two algorithms isn't required here (you can read the original publications for that), but know that Soundex always returns a four-character string where "0000" means that no letters were detected, and if Metaphone returns an empty string, then it too did not find any letters.

levenshtein()